

BUSINESS & TECHNICAL REPORT

Streaming IP Acquisition Screening Model

Predicting Community Momentum and Identifying Above-Expected Engagement to Prioritize Anime IP for Licensing Investigation

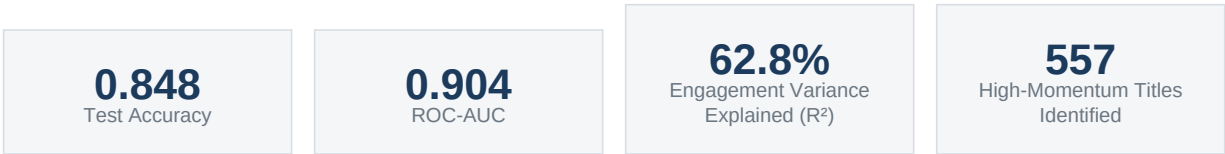
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01 Executive Summary

This report evaluates whether observable MyAnimeList audience and popularity signals can support a streaming platform's early-stage IP screening process. The analysis combines data quality validation, feature engineering, exploratory analysis, supervised classification, and a secondary above-expected engagement (residual) analysis to produce a ranked shortlist of titles warranting deeper commercial review.



The final classification model — Logistic Regression — achieved 0.848 accuracy, 0.705 precision, 0.661 recall, 0.682 F1, and 0.904 ROC-AUC on a held-out test set, outperforming both a majority-class baseline and a tested Random Forest alternative. A companion Linear Regression model explained 62.8% of the variation in log review count, enabling a residual-based ranking of titles whose community engagement exceeds what their audience size, popularity rank, and score would predict.

BOTTOM LINE FOR DECISION-MAKERS

*This model is a **screening signal**, not an acquisition verdict. It is designed to shrink a catalog of thousands of titles to a shortlist worth human review — licensing availability, cost, regional demand, demographics, competitive landscape, and content suitability are explicitly out of scope and must be layered on by business, legal, and content teams before any acquisition decision is made.*

02 Business Problem & Objectives

Streaming platforms face a continuous portfolio-selection problem: far more candidate titles exist than there is budget, catalog space, or licensing bandwidth to pursue. Without a structured first-pass filter, acquisition teams default to chasing already-famous titles or relying on ad hoc intuition — both of which risk overpaying for saturated IP or missing under-the-radar titles with real audience traction.

Primary Question

Can existing audience size, popularity rank, and rating signals identify anime with strong community momentum, well ahead of a full licensing review?

Secondary Question

Among titles with comparable audience and popularity characteristics, which ones are generating meaningfully more community engagement than expected — the titles a purely popularity-driven scan would overlook?

Intended Use

- Produce a **ranked screening shortlist**, not an automated acquisition decision.
- Give acquisition analysts a defensible, data-driven starting point for prioritizing deeper research.
- Surface engagement outliers that popularity rank alone would not reveal.

03 Data & Scope

The analysis uses the Kaggle “Anime Dataset with Reviews – MyAnimeList” dataset (Marlesson), combining anime metadata with user review records. Profile-level data was available but not required for this stage of the analysis.

Dataset / Stage	Result
Original anime records	19,311
Unique anime after UID deduplication	16,216
Original review records	192,112
Unique reviews after UID deduplication	130,519
Anime with ≥ 10 reviews (modeling population)	2,266
High-momentum threshold	>50 reviews (75th percentile)
High-momentum observations	557 (24.6%)

Scope boundary: this is a cross-sectional snapshot of MyAnimeList community activity. It reflects engagement among MyAnimeList users specifically, not verified streaming viewership, and carries no time-series component — a constraint revisited in Section 10.

04 Methodology & Data Quality

A structured verification pass preceded any modeling work: schema checks, missingness review, distribution inspection, duplicate-identifier resolution, and review-to-anime referential integrity testing. Anime records contained duplicate UIDs and were reduced to one record per UID; review records contained exact duplicate UID rows and were likewise deduplicated. Every retained review matched a valid anime UID post-cleaning.

Feature Selection & Leakage Control

Review-derived predictors (sentiment, review-text variability, etc.) were deliberately excluded from the final classification feature set. Because the modeling target is itself defined from review volume, including additional review-derived features would make the business question circular — the model would effectively be predicting the label from features derived from the label. The final classification predictors were restricted to **members**, **popularity**, and **score**.

Pipeline Overview

- Data Verification — dedup, missingness, schema and referential integrity checks
- Community Feature Engineering — review-volume metrics, log transforms
- Exploratory Data Analysis — correlation structure, skew handling, sentiment-vs-volume decoupling
- Classification Modeling — Logistic Regression vs. Random Forest vs. majority baseline
- Above-Expected Engagement Analysis — Linear Regression → residual-based ranking

05 Exploratory Findings

Review volume is heavily right-skewed — most anime accumulate only a handful of reviews while a small number accumulate hundreds or more — so log-transformed review count was used throughout the engagement modeling.

FIGURE 1 | CORRELATION STRUCTURE

Original chart asset was not available in the supplied environment; the reported analytical findings are retained below.

Community review volume is strongly associated with existing audience size (raw review_count ↔ members, $r \approx 0.828$). Log review count is also strongly associated with MAL popularity rank ($r \approx -0.768$) and positively associated with members ($r \approx 0.648$); since a *lower* MAL popularity rank denotes greater popularity, the negative correlation is directionally expected rather than anomalous.

Critically, review volume is **not a proxy for sentiment**. Among the most-reviewed titles in the dataset were both critically acclaimed works (e.g., *Death Note*, *Steins;Gate*) and titles with substantially lower average ratings (e.g., *School Days*). This supports treating engagement volume and audience sentiment as separate constructs rather than conflating “popular” with “well-reviewed.”

06 Classification Model: Predicting High Momentum

The modeling population was restricted to anime with at least 10 reviews (2,266 titles). Within that population, the 75th percentile of review count was 50 reviews; titles exceeding that threshold were labeled **high_momentum = 1** (557 titles), with the remainder labeled 0 (1,709 titles). The data was split 80/20 with stratification and a fixed random seed to ensure reproducibility.

A majority-class DummyClassifier established the baseline. Logistic Regression and Random Forest were then evaluated on identical held-out test data using accuracy, precision, recall, F1, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Majority Baseline	0.753	—	—	—	—
Random Forest	0.813	0.596	0.750	0.664	0.877
Logistic Regression — Final	0.848	0.705	0.661	0.682	0.904

FIGURE 2 | CLASSIFICATION PERFORMANCE

Original chart asset was not available; the held-out metrics are shown in the table above.

Logistic Regression outperformed both the baseline and Random Forest on every metric that matters most for a screening use case — ROC-AUC and F1 — while remaining fully interpretable, a meaningful advantage when a business stakeholder needs to understand *why* a title was flagged.

07 Model Interpretation

FIGURE 3 | STANDARDIZED COEFFICIENTS

Original coefficient chart asset was not available; the reported fitted coefficients are summarized in the narrative.

The **popularity** coefficient (-3.195) dominates the fitted model — consistent with EDA showing a strong relationship between popularity rank and review engagement. **Members** (+0.358) contributes a modest positive effect, and **score** (-0.454) contributes a smaller conditional effect.

INTERPRETATION CAUTION

*The negative **score** coefficient should not be read causally as “higher-rated anime get fewer reviews.” It reflects the model’s conditional relationship after accounting for popularity and members — a standard and expected behavior in multivariate regression, not evidence that quality suppresses engagement.*

08 Above-Expected Engagement Analysis

Classification answers a binary question: does a title belong to the high-momentum group? To add a more actionable screening layer, a Linear Regression model predicted log review count from members, popularity, and score — then compared each title's *actual* engagement to its *predicted* engagement. The difference is the **momentum residual**: positive values indicate a title is generating more community engagement than its observable audience and popularity signals would predict.

0.413
MAE

0.523
RMSE

0.628
R² (Test Set)

FIGURE 4 | TOP POSITIVE RESIDUALS

Original residual chart asset was not available; the reported ranked shortlist is reproduced below.

Title	Members	MAL Popularity	Reviews
Princess Tutu	110,983	1,010	94
Akatsuki no Yona	441,757	174	239
Kaze ga Tsuyoku Fuiteiru	118,998	955	84
Yuru Camp■	209,936	521	122
Mawaru Penguindrum	221,240	482	127
Mo Dao Zu Shi	76,350	1,418	60
Lovely★Complex	356,826	259	176
Phantom: Requiem for the Phantom	235,408	436	127
Cross Game	86,377	1,252	65

CONTENT-SUITABILITY SCREEN REQUIRED BEFORE ANY SHORTLIST SHIPS

One algorithmically-surfaced title in the raw above-expected engagement output was excluded from this shortlist following a content-suitability review — it is adult-oriented content unsuitable for general acquisition consideration. A purely statistical engagement signal has no awareness of content suitability, brand safety, audience appropriateness, or licensing eligibility. **Every output of this model must pass through human content review before reaching an acquisition conversation.**

09 Business Recommendations

The following recommendations translate model output into an operational workflow for an acquisitions team, rather than treating the model as a standalone decision system.

Use as a screening layer, not a decision system

Deploy community momentum and above-expected engagement scores as an additional filter early in IP research — never as a standalone acquisition criterion.

Prioritize above-expected engagement for deep-dive research

Titles with strong positive residuals should be routed to licensing, audience-overlap, regional-demand, and competitive-availability review ahead of titles that only rank high on raw popularity.

Pair every shortlist with a mandatory content-suitability gate

Statistical engagement signals can surface content that is unsuitable for acquisition regardless of its engagement score. No title should reach a licensing conversation without passing content and brand-safety review first.

Keep the classifier interpretable

Retain Logistic Regression as the production model for this use case; its transparent coefficients let analysts explain why a title was or wasn't flagged.

Treat the residual shortlist as a discovery tool, not a popularity leaderboard

Its purpose is to surface titles a pure popularity scan would overlook — measure its value by how many non-obvious titles convert into successful licensing deals.

10 Risks, Limitations & Governance

Responsible use of this model requires clear visibility into what it does *not* establish.

Limitation	Business Implication
Cross-sectional MyAnimeList snapshot, not longitudinal streaming data	Model measures association, not causal or forward-looking streaming performance.
High-momentum target is defined from review volume	The model identifies community engagement, not confirmed streaming success or profitability.
Members and popularity are contemporaneous platform metrics	These are naturally correlated with engagement by construction; treat coefficients as descriptive, not causal.
No licensing, cost, or rights data modeled	A positive screening score carries no information about acquisition feasibility or cost.
No demographic, regional, or competitive data modeled	Regional demand and audience-fit analysis must be performed separately before any decision.
Content suitability not modeled	Mandatory human content review is required before any shortlist reaches acquisition teams.
Dataset reflects MyAnimeList users specifically	Not a perfect representation of a platform's actual target streaming audience.

11 Roadmap & Next Iteration

The following extensions would move this from a screening prototype toward an operational acquisitions tool.

- **Longitudinal validation:** test whether community momentum precedes later popularity or confirmed streaming performance, rather than relying on a single snapshot.
- **Commercial data integration:** incorporate licensing availability, regional demand, watch-time / completion data, and acquisition economics.
- **Controlled feature expansion:** add genre and demographic features through an incremental, ablation-tested modeling process.
- **Threshold calibration:** tune classification thresholds against the real business cost of false positives versus missed opportunities.
- **Independent validation:** re-validate the full screening framework on a newer, independently collected dataset before relying on it operationally.

12 Appendix: Metrics & Technical Assets

Final Classification Metrics

Accuracy 0.848 | Precision 0.705 | Recall 0.661 | F1 0.682 | ROC-AUC 0.904

Above-Expected Engagement Regression Metrics

MAE 0.413 | RMSE 0.523 | R^2 0.628

Project Assets

- 00_Data_Verification.ipynb — data quality, duplicates, missingness, and referential-integrity validation.
- 01_Community_Features.ipynb — review and community feature engineering.
- 02_EDA.ipynb — engagement, rating, correlation, and genre analysis.
- 03_ML_Modeling.ipynb — classification, model comparison, regression, and above-expected engagement screening.
- outputs/models/ and outputs/tables/ — saved model artifacts and exported result tables.

Prepared using MyAnimeList community data (Kaggle: "Anime Dataset with Reviews", Marlesson). This document is intended for internal content-strategy review and should be read alongside the limitations in Section 10 before any business action is taken.